

Module 1: Cybersecurity Overview



Topic 4: Hash Cracking

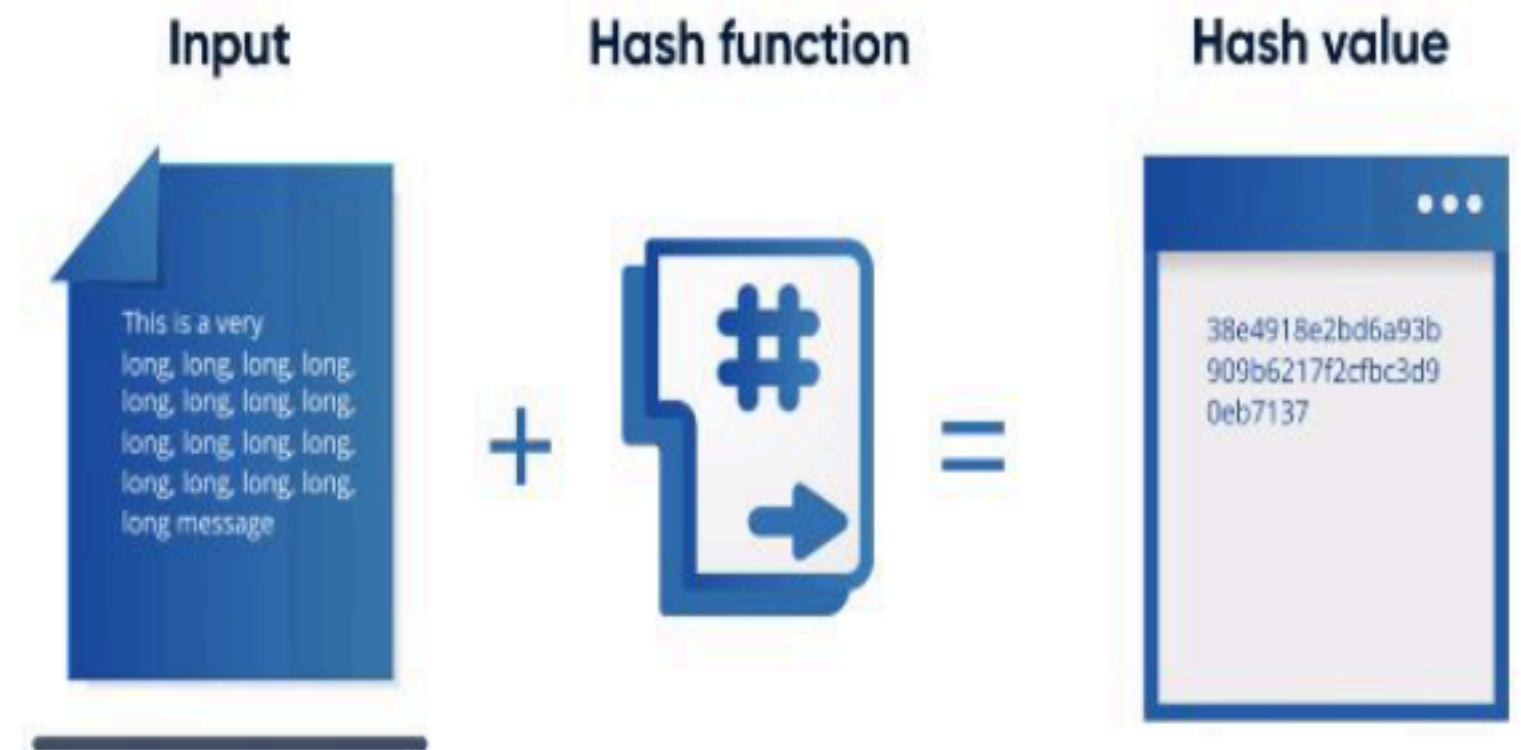


What is Hashing?

Hashing is one way algorithm used to convert data from its original format to unique, fixed-length string.

The main attributes for Hashing are:

- Unique
- Irreversible to original value (Secure)



Applications: Cryptocurrency, Digital signature, Malware identification, Evidence validation

Hashing Types

Each type of hashing should have its own length

MD5	128 bits / 32 characters
SHA1	160 bits / 40 characters
SHA2 (SHA256)	256 bits / 64 characters
CRC32	32 bits / 8 characters

Hashing Types

The different hash value for “Hello” word

MD5	8b1a9953c4611296a827abf8c47804d7
SHA1	f7ff9e8b7bb2e09b70935a5d785e0cc5d9d0abf0
SHA2 (SHA256)	185f8db32271fe25f561a6fc938b2e264306ec304eda518007d1764826381969
CRC32	9264541a

<https://md5calc.com/hash/crc32/Hello>

Hash Cracking

Although Hash algorithm should be secure by design.

The scientists are continuously challenging the security of the Hash by trying to find flaws in hashing algorithm to be reversible or not unique

It was found that MD5 and SHA1 is no longer secure because of collision attacks

Hash Cracking

The original value of a hash can be retrieved through some techniques such as:

- Dictionary attacks
- Brute Force attacks
- Rainbow tables attacks

How to Crack a Hash?

Hash Cracking Websites

<https://crackstation.net/>

<https://www.md5online.org/>

<https://hashkiller.io/>

How to Crack a Hash?

John The Ripper Tool

wordlist

Bruteforce



How to Crack a Hash?

Hashcat

wordlist

Bruteforce



Lab

Guess the Password

